

A Synopsis

On

Food Delivery App System

Submitted to

IILM UNIVERSITY

In Partial Fulfilment for the award of the degree of

B.TECH

In

COMPUTER SCIENCE AND ENGINEERING

By

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March 2026

PROJECT DETAILS

TITLE OF THE RESEARCH PROPOSAL	Food Delivery App System
NAME OF THE STUDENTS/GROUP	Muskan kumari
NAME AND DESIGNATION OF THE GUIDE	Ms. Parul Saini Associate Professor
PLACE OF WORK	IILM University, Greater Noida
AREA	Web Technology / Full Stack Development
DOMAIN	Web Application
CO-DOMAIN	Database Management, Cloud Computing
APPLICATION AREA	Online Food Ordering for Small Restaurants, Street Food Vendors, Home-Based Kitchens
USEFULNESS	Provides a low-cost, simple, and local-friendly platform for small food businesses to go digital
ANY OTHER KEY INFORMATION	Low commission model (5-10%), multi-language support (Hindi/English), real-time tracking, manual order acceptance

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Statement of the proposal

The proposed project is a **web-based food delivery application** designed specifically for **small restaurants, street food vendors, and home-based kitchens**. It provides a simple, affordable, and user-friendly platform to help local food businesses go digital and compete with large food delivery platforms.

Problem Statement

In today's fast-paced life, people prefer ordering food online rather than dining out. Existing food delivery platforms like Zomato and Swiggy charge high commissions (25-30%), which small restaurants and local vendors cannot afford. As a result, these businesses remain offline and lose potential customers. There is a need for a **low-cost, simple, and local-friendly food delivery system** that empowers small food businesses.

Background

With the rapid growth of online food delivery, large platforms have dominated the market. However, small restaurants and street food vendors are left behind due to high costs and complex technology. This project aims to bridge that gap by providing a platform that is **affordable, easy to use, and focused on local needs**.

Motivation/need for the research:

- Increasing demand for online food services
- Lack of customized platforms for small-scale restaurants
- Need for real-time tracking and seamless user experience
- Opportunity to integrate AI-based recommendations and analytics
- High commission rates of existing platforms exclude small businesses

Objective

To design and develop a fully functional **Food Delivery Web Application** that provides a seamless experience for users, restaurants, and delivery partners. **Application** that enables small restaurants and local food vendors to accept online orders with **low commission, simple interface, and multi-language support**.

Sub-Objectives

1. User authentication and profile management
2. Restaurant and menu management system
3. Provide free registration for first 3 months to small vendors
4. Keep commission low (5-10%)
5. Design a simple dashboard with Hindi/English language option
6. Allow manual order acceptance for restaurants
7. Integrate real-time order tracking for customers
8. Include multiple payment options (UPI, Cash, Card)
9. Provide rating and review system for local food

Mode of achieving objective

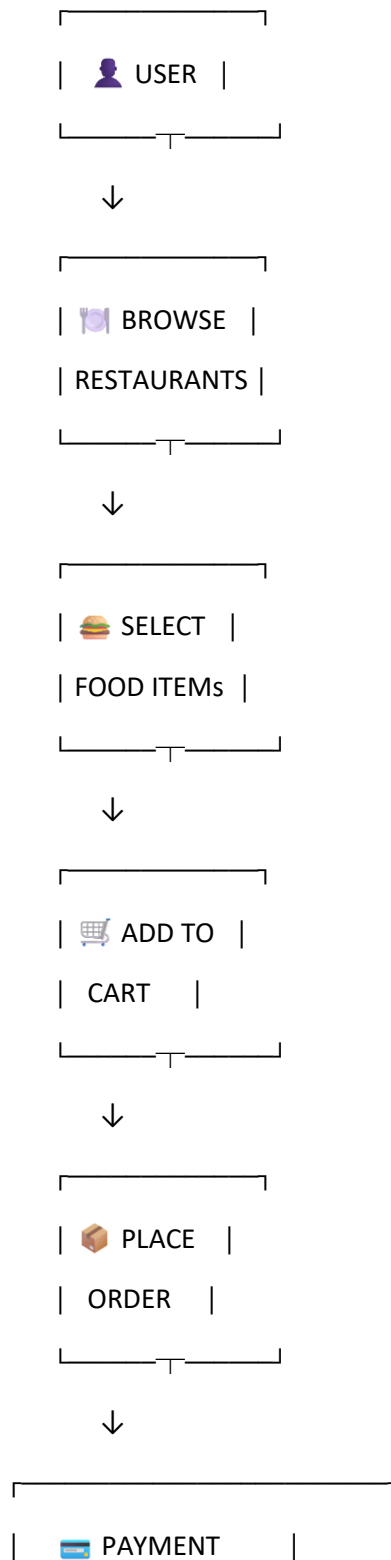
1. Requirement gathering from small restaurant owners
2. System design with focus on simplicity
3. Frontend development using HTML, CSS, JavaScript (React/Angular)
4. Backend development using Node.js / PHP / Python
5. Database design using MySQL / MongoDB
6. Integration of APIs for maps and payments
7. Testing and deployment Proposed Methodology

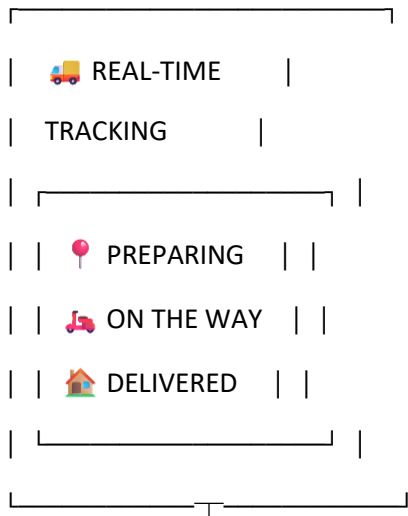
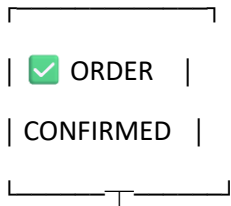
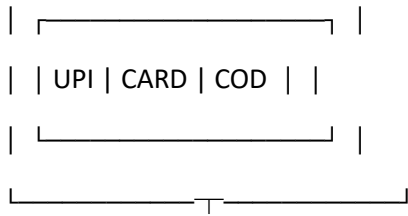
Theoretical framework – The system will follow **Client-Server Architecture** with MVC pattern. The frontend will interact with RESTful APIs to communicate with the backend and database. Special attention will be given to **simplicity and multi-language support**.

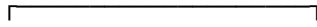
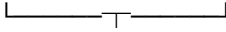
Sources of data

- **Primary data:** Interviews with small restaurant owners, street food vendors, and customers
- **Secondary data:** Research papers, case studies on Zomato, Swiggy, and local food businesses

Figure 1. Work flow Diagram (Clarity on developing System/Method/Model/Platform/Apparatus)







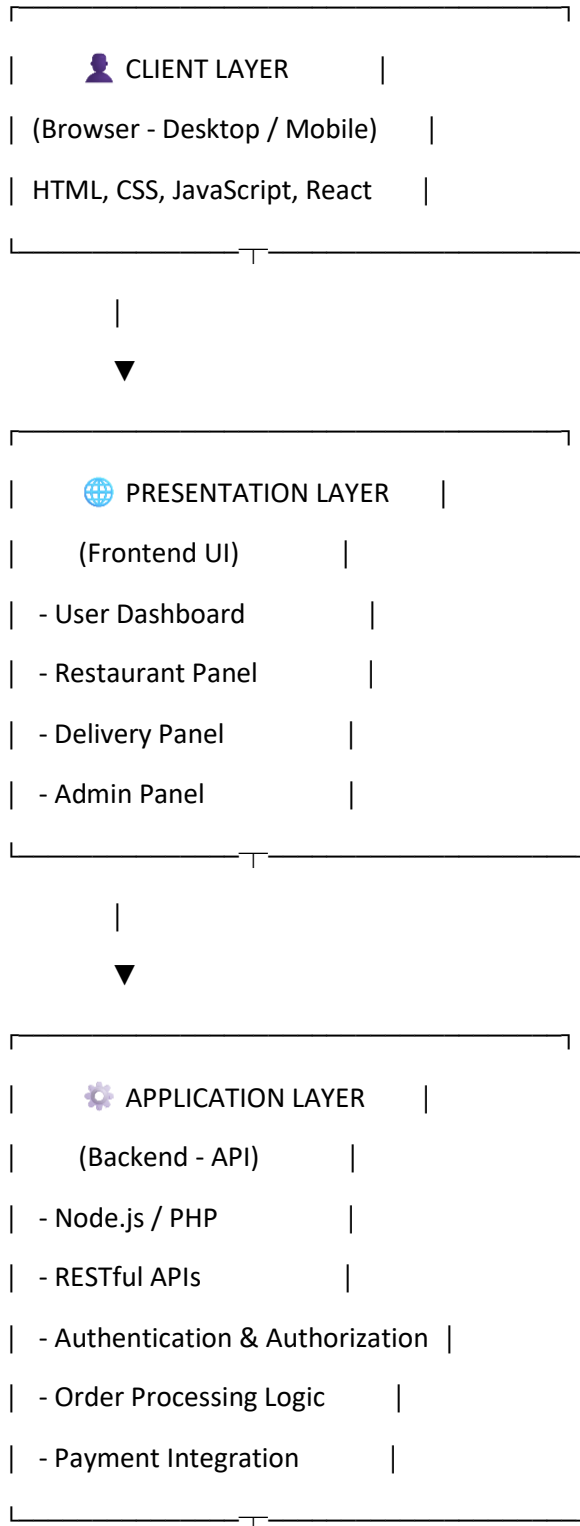
| BACK TO USER |

| (Circular Flow) |



Schematic flow Diagram:

Figure 2. Schematic Flow Diagram



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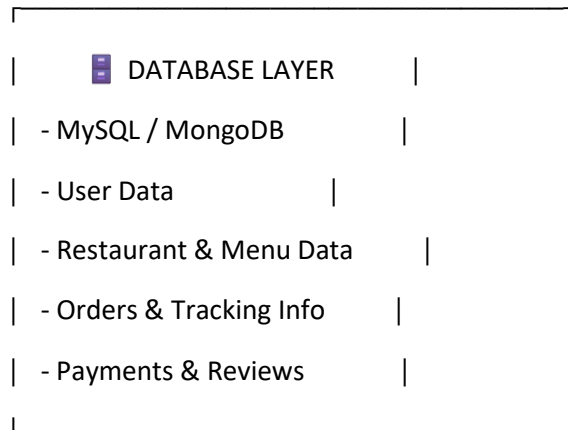


Figure 3. Gantt chart

PROJECT TIMELINE (Weeks 1 to 12)



Review of literature

Essential Literatures reviewed	Themes drawn from the Literatures
Online Food Delivery Systems – A Review	User behavior, UI/UX design
Real-time Tracking in Food Apps	GPS integration, latency issues
Payment Gateway Integration	Security, UPI, Cards, Wallet

Research Gaps:

Essential Literatures reviewed	Themes drawn	Research Gaps	Remark
Swiggy Tech Blog	Scalability	Lack of small restaurant focus	Need local vendor inclusion
Zomato Case Study	Personalization	AI recommendations not deep	Add AI-based suggestions

Key Bibliography

1. Pressman, R. (2014). Software Engineering: A Practitioner's Approach
2. Zomato & Swiggy Annual Reports & Case Studies
3. Research Papers on Online Food Delivery Systems (IEEE, Springer, Elsevier)
4. MDN Web Docs – Web Development Guides
5. Razorpay & Stripe Payment Gateway Documentation
6. Google Maps API Documentation
7. "Impact of Food Delivery Apps on Small Restaurants" – Journal of Food Service Technology

